

- 1) Define Data Mining and list any two basic data mining tasks. Explain the steps involved in KDD process and justify the role of data preprocessing in it.

Data Mining: Finding useful patterns from large data.

Tasks: Classification, clustering.

KDD steps: Cleaning → Integration → selection → Transformation →
Data mining → Evaluation → Presentation.

Role of preprocessing: Improves data quality and accuracy.

- 2) What is data cleaning? Mention two common data quality issues. Explain methods to handle missing values and noisy data with suitable examples.

Data cleaning: Removes errors and missing data.

Issues: Missing values, Noisy data.

Methods: Replace missing values (mean/mode), smooth noisy data (binning).

- 3) Define data integration and list two challenges associated with it. Explain how redundancy and inconsistency are handled during data integration.

Definition:- Combining data from different sources.

Challenges: Redundancy, Inconsistency.

Solution:- Remove duplicates and standardize formats.

4) What is covariance ~~system~~ analysis? List the two covariance forms.

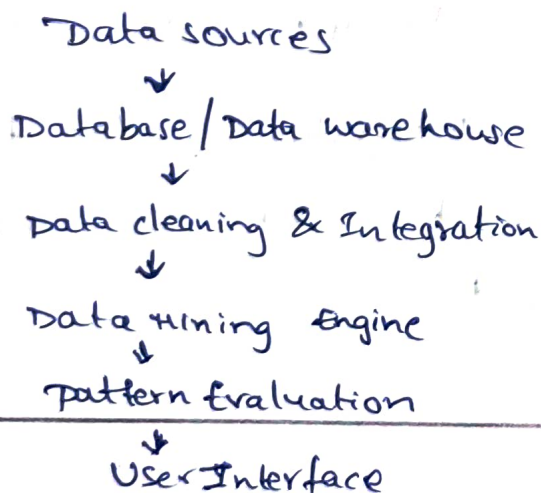
Definition: Measures how two variables change together.

Forms: Population Covariance, sample Covariance.

5) Suppose two stocks A and B have the following values in one week: (2,5); (3,8), (5,10), (4,11), (6,14). If the stocks are affected by the same industry trends, will their prices rise or fall together and independent?

Positive Covariance Both stocks rise together.

6) Explain the architecture of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support Knowledge discovery.



7) find covariance for following data set
 $x = \{4, 5, 6, 7, 9\}$, $y = \{8, 3, 7, 5, 6\}$

Given

$$x = \{4, 5, 6, 7, 9\}, y = \{8, 3, 7, 5, 6\}$$

$$\text{Mean}(x) = 6.2, \text{Mean}(y) = 5.8$$

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = -2.8$$

$$\text{Sample Covariance} = \frac{-2.8}{5-1} = -0.7$$

Covariance = -0.7 (negative relationship)

8) Describe the major steps in data preprocessing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps.

- Cleaning $\rightarrow$ Integration $\rightarrow$ Transformation $\rightarrow$ Reduction
 $\rightarrow$ Discretization

Pipeline: Fix missing values $\rightarrow$ Standardize $\rightarrow$ Remove outliers $\rightarrow$ Normalize.

9) A hospital is analyzing Data
Equal-frequency Binning
3 Bins (9 values each):

• Bin1: 13-22

• Bin2: 22-35

• Bin3: 35-76

10) Data Discretization

Definition: converts continuous data into intervals

Importance: Reduces noise and simplifies analysis

Equal-width: same interval size.

Equal-Frequency: - same number of values per bin.

cheer!